Distinct running least common multiples

Affine tails and Erdős Problem #269

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Abstract

We give a five-map proof that the reciprocals of the distinct running least common multiples of the 5-smooth integers have an irrational sum. The spacing of powers of 5 separates the affine tail images; rationality would then force periodicity, contrary to the frequency of powers of 3. For the sum counted with multiplicity, the tails instead give an integer recurrence and an equivalent residue criterion for irrationality. Proving that criterion for the actual coefficients remains open here.

1 Two sums from running least common multiples

Let P be a finite set of primes and $\mathcal{S}_P$ the positive integers with prime factors in P, including 1. For $P = \{2, 3, 5\}$, the running least common multiples of these 5-smooth integers begin $1, 2, 6, 12, 60, 60, 120, 360, 360, \ldots$. Put

$$\mathrm{L}(x) = \operatorname{lcm}\{u \in \mathcal{S}_P : u \le x\}, \qquad \mathrm{H}_P(x) = \prod_{p \in P} p^{\lfloor \log_p x \rfloor} \quad (x \ge 1).$$

Every member of the prefix divides $\mathrm{H}_P(x)$, while the prefix contains the largest power of each $p \in P$ at most x. Thus $\mathrm{L}(x) = \mathrm{H}_P(x)$, with a jump by p exactly at each positive power of p. Counting distinct running LCMs or retaining multiplicities gives

$$\mathcal{D}_P = 1 + \sum_{\substack{t = p^n \\ p \in P,\ n \ge 1}} \frac{1}{\mathrm{H}_P(t)}, \qquad \mathcal{R}_P = \sum_{u \in \mathcal{S}_P} \frac{1}{\mathrm{L}(u)}.$$

For $P = \{2, 3, 5\}$, $\mathrm{L}(5) = \mathrm{L}(6) = 60$ contributes $1/60$ to $\mathcal{D}_P$ and $2/60$ to $\mathcal{R}_P$. Erdős Problem #269 asks about $\mathcal{R}_P$ [1, p. 65] and [2, p. 106]. Positive powers of distinct primes never coincide; geometric growth of successive running LCMs gives convergence of $\mathcal{D}_P$. We use $\mathbb{N} = \{0, 1, 2, \ldots\}$.

Theorem 1.1 (the sum over distinct running LCMs for $\{2, 3, 5\}$)**.** *The number $\mathcal{D}_{\{2,3,5\}}$ is irrational.*

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Rationality would leave only finitely many normalised tails, but finiteness alone does not force the jumps to repeat. The decisive step is to recover the next block and the following tail from the current tail. Powers of 2 delimit the blocks; the spacing of powers of 5 separates their five affine images. Equal tails then have identical futures, contradicting the irrational frequency of powers of 3.

Erdős asserted irrationality of $\mathcal{D}_P$ for every finite P with $|P| \geq 2$ in a letter dated 1 January 1973, without printing a proof [3, p. 335]. Thus the theorem above is a special case of that earlier assertion. We claim no priority for its conclusion. The companion's general argument uses an isolated interchange of jumps on a torus. That argument is retained there as an ordinary proof awaiting formalisation. The proof below is independent of that argument.

Our arguments leave the repeated sum $\mathcal{R}_P$ unresolved for finite P with $|P| \geq 3$. For $P = \{p\}$ both sums equal $p/(p-1)$.

After the five-map proof in Section 2, we retain the multiplicities. The bases remain the same, but the block coefficients and tails grow quadratically. Sections 3 and 4 replace finite-state repetition by a test for excluding integer tails. For two primes, Fan's factorisation [7] reduces both sums to a Hecke–Mahler value and yields transcendence (Section 5). The appendices supply the running LCM identities and show why exact finite separation of the three-prime kernel cannot extend that factorisation.

2 Recovering the prime-power jumps

Long division combines a finite set of remainders with a deterministic rule for the next digit and remainder. Here the bases vary with the prime-power jumps, so that rule must be proved. We first obtain finitely many tails under rationality, then recover the jumps from each tail.

Proof of Theorem 1.1. Finitely many tails under rationality. Write $\mathrm{H} = \mathrm{H}_{\{2,3,5\}}$ in this proof and put $P_a = \mathrm{H}(2^a)$. Every running LCM at or before 2^a divides P_a. We therefore normalise the tail after 2^a by

$$Y_a = P_a \sum_{\substack{t>2^a \\ t=2^n,3^n \text{ or } 5^n,\ n\geq 1}} \frac{1}{\mathrm{H}(t)}. \tag{1}$$

Each subsequent jump multiplies the running LCM by at least 2, and a jump by 3 eventually occurs. Comparison with $\sum_{j\geq 1} 2^{-j}$ gives $0 < Y_a < 1$.

Suppose that $\mathcal{D}_{\{2,3,5\}} = N/K$, where N is an integer and $K \geq 1$ is an integer. Clearing the finite prefix gives

$$KY_a = P_a N - K\left(P_a + \sum_{\substack{t\leq 2^a \\ t=2^n,3^n \text{ or } 5^n,\ n\geq 1}} \frac{P_a}{\mathrm{H}(t)}\right) \in \mathbb{Z}.$$

Thus the Y_a lie in the finite set $(K^{-1}\mathbb{Z}) \cap (0,1)$. It remains to show why equality of two tails propagates.

Recovering a block from its tail. Consider the jumps between 16 and 32, including the last endpoint. They occur at $25, 27, 32$, multiplying the running LCM successively by $5, 3, 2$. After normalisation at 16, their contribution is $1/5 + 1/15 + 1/30$. The remaining tail is normalised at a running LCM 30 times as large, so

$$Y_4 = \frac{1}{5} + \frac{1}{15} + \frac{1}{30} + \frac{Y_5}{30} = \frac{9 + Y_5}{30}.$$

In the block $(64, 128]$, the interior jumps occur in the opposite order, at $81, 125$. Its constant term is $1/3 + 1/15 + 1/30 = 13/30$. The multiplier 30 records which primes occur, but not their order. The numerators 9 and 13 retain that order. To read it from a tail, we must also control the unknown following tail.

Every block $(2^a, 2^{a+1}]$ contains its terminal power of 2, at most one power of 3 and at most one power of 5. We record the interior jumps in their order by a word τ_a on the letters $3, 5$, omitting the terminal 2. Splitting off the block gives

$$Y_a = G_{\tau_a}(Y_{a+1}), \qquad G_\tau(y) = \frac{\mu_\tau + y}{b_\tau}, \qquad \begin{array}{c|ccccc} \tau & \emptyset & 3 & 5 & 35 & 53 \\ \hline b_\tau & 2 & 6 & 10 & 30 & 30 \\ \mu_\tau & 1 & 3 & 3 & 13 & 9 \end{array} \tag{2}$$

Here b_τ is the factor by which the running LCM increases across the block, and μ_τ / b_τ is the block's normalised contribution.

The bound $0 < Y_{a+1} < 1$ does not yet recover τ_a: $G_{53}([0,1]) = [3/10, 1/3]$ and $G_5([0,1]) = [3/10, 2/5]$ overlap. The table improves the lower bound to $Y_a \geq 3/10$. To separate these two images, an upper bound U must satisfy

$$G_{53}(U) < G_5(3/10), \qquad \frac{9 + U}{30} < \frac{33}{100}, \qquad U < \frac{9}{10}.$$

The spacing of powers of 5 supplies such a bound. A power 5^f belongs to block $\lfloor f \log_2 5 \rfloor$. These indices start at 2 and their successive differences are 2 or 3. Every three consecutive blocks therefore contain a power of 5. Given a, let $j \in \{a, a+1, a+2\}$ be the first such block. The maps for blocks containing 5 give $Y_j \leq 7/15$, since $Y_{j+1} < 1$ and $G_{35}(1) = 7/15$ bounds $G_5(1)$ and $G_{53}(1)$ as well. We now work *backwards* from j to a. Each intervening block has map $G_\emptyset$ or G_3, and $G_3(y) \leq G_\emptyset(y) = (1 + y)/2$ for $y \geq 0$. At most two backward steps give the successive bounds $7/15, 11/15, 13/15$. Hence

$$Y_a \in I = \left[\frac{3}{10}, \frac{13}{15}\right], \qquad G_\emptyset(G_\emptyset(7/15)) = \frac{13}{15} < \frac{9}{10}. \tag{3}$$

This bound holds for the tails of the prime-power sequence because every three consecutive blocks contain a power of 5. The interval I is not invariant under arbitrary compositions of the five maps: for example, $G_\emptyset(13/15) = 14/15$ lies outside I.

Figure 1 gives the exact images of I. They are pairwise disjoint, with successive gaps $1/900, 17/300, 79/900, 1/180$. Writing $L = 3/10$ and $U = 13/15$, the two narrow gaps are

$$G_5(L) - G_{53}(U) = \frac{3L - U}{30} = \frac{1}{900}, \qquad G_\emptyset(L) - G_3(U) = \frac{3L - U}{6} = \frac{1}{180}.$$

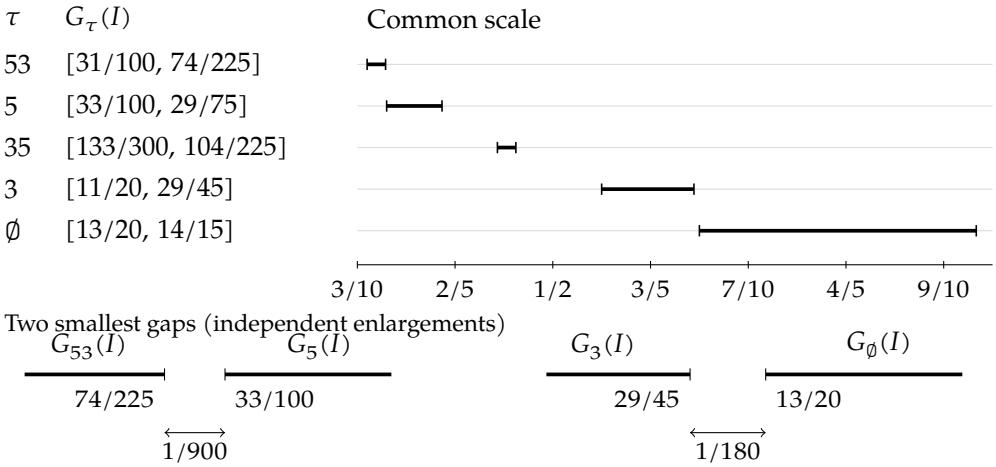


Figure 1: The five images of the arithmetic tail interval $I = [3/10, 13/15]$. Their separation determines the next block from an actual tail. The full image $G_\emptyset(I)$ extends beyond I; the figure does not assert invariance under arbitrary words. Only the adjacent endpoints are shown in the two lower enlargements.

Thus the same inequality $U < 3L = 9/10$ separates both pairs.

Each actual Y_a lies in exactly one image interval, identifying τ_a and then $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$. For instance, $Y_a \in [31/100, 74/225]$ forces $\tau_a = 53$ and $Y_{a+1} = 30Y_a - 9$. This is the next arithmetic tail, not an arbitrary inverse image; it lies in I, so recovery can be repeated.

Repetition would give a rational frequency. The finitely many tail values obtained above include a repeated value $Y_u = Y_v$ with $u < v$. Since the image intervals are disjoint, this forces $\tau_u = \tau_v$. Applying the corresponding inverse map gives $Y_{u+1} = Y_{v+1}$. Induction yields $\tau_{u+j} = \tau_{v+j}$ for every $j \geq 0$. Thus the entire later block word has period $v - u$.

To exclude this possibility, let $\alpha = \log 2 / \log 3$. The number of powers of 3 in block a is $\lfloor (a+1)\alpha \rfloor - \lfloor a\alpha \rfloor$, and is determined by τ_a. If the block word has eventual period $\ell \geq 1$, let c be the number of 3-powers in one period. For every sufficiently large fixed a and every $n \geq 1$ we then have

$$\lfloor (a + n\ell)\alpha \rfloor = \lfloor a\alpha \rfloor + nc.$$

Dividing by n and taking the limit gives $\ell\alpha = c$, contrary to unique factorisation, since it would imply $2^\ell = 3^c$. □

3 The repeated three-prime sum

We now retain every smooth integer, not just the jumps of its running LCM. Put $P = \{2, 3, 5\}$, $S = \mathcal{R}_P$ and $\mathrm{H} = \mathrm{H}_P$ throughout this section. The denominator changes at the same prime powers; the number of terms at each denominator is what changes the argument.

For example, $[16, 32)$ contains the seven smooth integers $16, 18, 20, 24, 25, 27, 30$. Grouping them by their running LCM gives

smooth integers	running LCM	contribution
$16, 18, 20, 24$	720	$4/720$
25	3600	$1/3600$
$27, 30$	10800	$2/10800$

The distinct sum counts each displayed running LCM once; the repeated sum assigns the coefficients $4, 1, 2$. The jumps alone therefore no longer determine the block contribution. Appendix A gives the general finite grouping identity.

With blocks $[2^a, 2^{a+1})$, the tail now includes 2^a. The prefix excludes the jump by 2 there, so half the running LCM clears it. Put

$$s_a = \sum_{\substack{x \in \mathcal{S}_P \\ 2^a \le x < 2^{a+1}}} \frac{1}{\mathrm{H}(x)}, \qquad T_a = \sum_{j \ge 0} s_{a+j}, \qquad h_a = \frac{\mathrm{H}(2^a)}{2}, \qquad X_a = h_a T_a.$$

For $a \ge 1$, each prefix LCM has exponent of 2 at most $a - 1$ and divides the integer h_a; at $a = 0$, $h_0 = 1/2$. The factor $1/2$ reflects the endpoint change from Section 2, not multiplicity.

Splitting $T_a = s_a + T_{a+1}$ and using $h_{a+1} = b_a h_a$ gives $X_{a+1} = b_a X_a - m_a$, where $m_a = h_{a+1} s_a$. Explicitly,

$$b_a = \frac{\mathrm{H}(2^{a+1})}{\mathrm{H}(2^a)}, \qquad m_a = \sum_{\substack{x \in \mathcal{S}_P \\ 2^a \le x < 2^{a+1}}} \frac{\mathrm{H}(2^{a+1})}{2\,\mathrm{H}(x)}. \tag{4}$$

Since $h_0 = 1/2$, we also have $h_{a+1} = (b_0 \cdots b_a)/2$ and $m_a/(b_0 \cdots b_a) = s_a/2$. For $[2, 4)$, the running LCMs at $2, 3$ are $2, 6$ and $h_2 = 6$, giving $m_1 = 6(1/2 + 1/6) = 4$ and $b_1 = 6$. For the block displayed above, $h_5 = 10800$ and

$$m_4 = 4 \cdot 15 + 3 + 2 \cdot 1 = 65, \qquad b_4 = 30.$$

Lemma 3.1 (integer coefficients and four possible bases)**.** *For every $a \ge 0$, m_a is a positive integer and $b_a \in \{2, 6, 10, 30\}$. The word "numerator" does not impose the positional-digit restriction $m_a < b_a$; that restriction need not hold.*

Proof. For $x < 2^{a+1}$ we have $2\,\mathrm{H}(x) \mid \mathrm{H}(2^{a+1})$, so every summand in m_a is integral. The interval contains 2^a, giving positivity. Between successive powers of 2 there is at most one power of 3 and at most one power of 5. Along with the final factor 2, these give $b_a \in \{2, 6, 10, 30\}$. □

Proposition 3.2 (the tail recurrence and a quadratic bound)**.** *The series defining S, T_a converge. For every $a \ge 0$,*

$$X_{a+1} = b_a X_a - m_a, \qquad 0 < X_a \le \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \ge 1$, either BX_a is integral at some index (and hence at every later index), or $\mathrm{dist}(BX_a, \mathbb{Z}) \ge 1/31$ at arbitrarily large indices.

Proof. We count at most $(a+1)^2$ smooth integers in $[2^a, 2^{a+1})$: each pair of exponents of $3, 5$, both at most a, allows at most one exponent of 2. By (12), $s_a \le 30(a+1)^2/8^a$. As $h_a \le 8^a/2$ and $a+j+1 \le (a+1)(j+1)$, we have

$$X_a \le 15(a+1)^2 \sum_{j\ge 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This also proves convergence. Splitting off s_a gives the recurrence, since $h_{a+1} = b_a h_a$ and $m_a = h_{a+1} s_a$.

For the final assertion, suppose that every sufficiently late distance is strictly less than $1/31$, and write $BX_a = z_a + e_a$, where z_a is integral and $|e_a| < 1/31$. The recurrence makes $e_{a+1} - b_a e_a$ an integer of absolute value less than $(1+b_a)/31 \le 1$. Thus $e_{a+1} = b_a e_a$. A nonzero error would therefore at least double at each step, contradicting $|e_a| < 1/31$. All sufficiently late errors vanish; integrality then persists by the recurrence. □

The loss of bounded tails is genuine. For $a \ge 1$, every odd part $3^j 5^k < 2^a$ has a unique multiple by a power of 2 in $[2^a, 2^{a+1})$. Taking $0 \le j, k \le \lfloor a/5 \rfloor$ ensures this inequality because $15 < 2^5$. Each such integer contributes at least 1 to m_a, so

$$m_a \ge (\lfloor a/5 \rfloor + 1)^2, \qquad X_a > m_a/30.$$

Rationality would still put the tails in a fixed lattice, but not in a finite set. Repetition alone also gives no recovery rule: the maps $y \mapsto (m_a + y)/b_a$ now depend on a and form an infinite family. The proof in Section 2 needed both finiteness and unique recovery. We instead use the quadratic bound to limit the positive integer tails required by rationality.

The Cantor-series form $S/2 = \sum_{a\ge 0} m_a/(b_0 \cdots b_a)$ also explains why the criteria in [9, Theorem 2.1] and [10, Theorem 3.1] do not apply: their small-numerator hypothesis $m_a/(b_{a-1}b_a) \to 0$ fails with bounded b_a and these unbounded m_a. The companion's tail estimates give sharper bounds.

Theorem 3.3 (rationality gives positive integer tails)**.** *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \ge a_0 = u + 1 + 2v + 3w$,*

$$d_a = BX_a \in \mathbb{Z}_{>0}, \qquad d_{a+1} = b_a d_a - B m_a, \qquad d_a \le 90B(a+1)^2.$$

Proof. For $a \ge 1$, clearing the finite prefix gives

$$X_a = h_a S - \sum_{\substack{x\in\mathcal{S}_P \\ x<2^a}} \frac{h_a}{\mathrm{H}(x)}, \qquad \sum_{\substack{x\in\mathcal{S}_P \\ x<2^a}} \frac{h_a}{\mathrm{H}(x)} \in \mathbb{Z}. \tag{5}$$

If $a \ge u+1+2v+3w$, the exponent $a-1$ of 2 in h_a is at least u, and $2^a \ge 3^v, 5^w$. Therefore $2^u 3^v 5^w \mid h_a$, and BX_a differs from $h_a A/(2^u 3^v 5^w)$ by an integer. Positivity, recurrence and the bound follow from Proposition 3.2. □

Corollary 3.4 (the first index at which the denominator clears)**.** *Under the same lowest-terms hypothesis, put $M = 2^u 3^v 5^w$ and let* den *denote the positive reduced denominator. For $a \ge 1$,*

$$\mathrm{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Consequently BX_a is integral exactly when $2^a \geq \max(2^{u+1}, 3^v, 5^w)$. The first such a can be found by integer comparisons, without logarithmic rounding.

Proof. Equation (5) shows that BX_a and h_aA/M have the same reduced denominator. Since $\gcd(A, M) = 1$, this denominator is $M/\gcd(M, h_a)$. Divisibility by each of $2^u, 3^v, 5^w$ gives the three stated inequalities. □

The two denominator parts have different roles: M determines when clearing occurs; B is the multiplier left after clearing. For $D = 2^3 3^2 5 \cdot 7$, the first integral $7X_a$ would occur at $a = 4$, not the sufficient index 11 in the theorem. Thus an early nonintegral tail need not stay nonintegral. The companion's denominator calculation also gives the reduced denominator of X_a itself.

4 A residue criterion

A finite segment bounds the starting tail. Its integrality would constrain the endpoint tail to a residue class. The residue inequality expresses the same obstruction in either form.

For two steps the recurrence reads $X_{\ell+2} = b_{\ell+1}b_\ell X_\ell - (b_{\ell+1}m_\ell + m_{\ell+1})$. To keep track of the multiplicative and additive terms over h steps, write

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \qquad W_{\ell,h+1} = b_{\ell+h}W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h}F_{\ell,h} + m_{\ell+h}.$$

Induction yields

$$X_{\ell+h} = W_{\ell,h}X_\ell - F_{\ell,h}. \tag{6}$$

Since $b_a = h_{a+1}/h_a$, its product telescopes to $W_{\ell,h} = h_{\ell+h}/h_\ell$. Thus

$$\frac{F_{\ell,h}}{W_{\ell,h}} = h_\ell \sum_{a=\ell}^{\ell+h-1} s_a, \qquad \frac{X_{\ell+h}}{W_{\ell,h}} = h_\ell T_{\ell+h}.$$

These finite and remaining contributions sum to X_ℓ, both in the starting normalisation h_ℓ, by (6). Throughout this section let

$$K(B, a) = 90B(a+1)^2. \tag{7}$$

The bound $0 < BX_{\ell+h} \leq K(B, \ell+h)$ and (6) enclose the starting tail in the interval

$$\frac{BF_{\ell,h}}{W_{\ell,h}} < BX_\ell \leq \frac{BF_{\ell,h} + K(B, \ell+h)}{W_{\ell,h}}.$$

Width below 1 gives at most one integer, not necessarily none. With $W = W_{\ell,h}$ and $F = F_{\ell,h}$, the interval contains no integer exactly when

$$BF + K(B, \ell+h) < W(\lfloor BF/W \rfloor + 1).$$

The right side is W times the first integer strictly above BF/W. Its excess over BF is the least positive residue of $-BF$ modulo W: the residue measures the gap to that integer in units of $1/W$.

For integers $W \geq 1, t$, define $\operatorname{lpr}_W(t) = 1 + ((t-1) \bmod W)$, with remainder in $\{0, \ldots, W-1\}$. Its values lie in $\{1, \ldots, W\}$, so the zero class is represented by W. This matches the open left endpoint: if $W \mid BF$, the next possible integer is one full unit above BF/W, not BF/W itself.

Lemma 4.1 (least positive residues)**.** *If d is a positive integer with $d \leq K$ and $d \equiv -BF \pmod{W}$, where $W \geq 1$, then $\operatorname{lpr}_W(-BF) \leq K$.*

Proof. The least positive representative is at most every positive representative of its class, and in particular at most d. □

In (6), an integral BX_ℓ makes $BX_{\ell+h}$ such a positive representative.

For $\ell = 1, h = 6$, direct iteration gives $W_{1,6} = 648000$ and $F_{1,6} = 524431$. Since $K(1,7) = 5760$, the interval is

$$\frac{524431}{648000} < X_1 \leq \frac{530191}{648000} < 1.$$

Equivalently, $\operatorname{lpr}_{W_{1,6}}(-F_{1,6}) = 123569 > 5760$ certifies $X_1 \notin \mathbb{Z}$, treating only $B = 1$ at $\ell = 1$. For denominators MB, varying B covers the part coprime to 30; arbitrarily late starts allow each supported part M to clear.

Theorem 4.2 (a residue criterion for irrationality)**.** *The number S is irrational if and only if*

$$\begin{gathered} \text{for every } B \geq 1 \text{ with } \gcd(B,30) = 1 \text{ and every } a_0 \geq 1, \\ \text{there are } \ell \geq a_0,\ h \geq 1 \text{ such that } \operatorname{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell+h). \end{gathered} \tag{8}$$

Proof. Suppose that (8) holds and S is rational. By Theorem 3.3, $d_a = BX_a$ is a positive integer for every sufficiently large a. Choose a segment beyond that onset. Equation (6) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contrary to Lemma 4.1 and (8).

Conversely, suppose that S is irrational and fix $B, \ell \geq 1$. Equation (5) makes BX_ℓ nonintegral. The distance upwards to the next integer, $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0,1)$, is now fixed as the segment length h grows. We have $W_{\ell,h} \geq 2^h$, whereas $X_{\ell+h}$ has only quadratic growth. Consequently $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0,1)$ for sufficiently large h. This ensures that the following representative lies strictly between 0 and the modulus, so it is the least positive one:

$$\operatorname{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h} = \delta W_{\ell,h} + BX_{\ell+h}.$$

The positive constant δ multiplies an exponentially growing $W_{\ell,h}$, so the first term eventually exceeds $K(B, \ell+h)$. This proves the inequality from every fixed start, and hence after every prescribed onset a_0. No bound uniform in B or ℓ is needed for this direction. □

At a fixed start, success rules out integrality of BX_ℓ. If BX_ℓ is integral, the recurrence, positivity and the bound prevent success at any length; for all sufficiently large h, $\operatorname{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = BX_{\ell+h}$. If BX_ℓ is nonintegral, every sufficiently long segment succeeds, even when S is rational with a denominator not yet cleared at ℓ. Increasing the

length does not remove the need for starts beyond every cutoff. A finite failed search distinguishes neither case.

The converse obtains $\delta > 0$ from irrationality. Proving (8) without that assumption is the remaining task. The upper-bound property of K is essential: a zero bound would make every residue inequality automatic. The companion's residue analysis treats the fixed-start formula and smaller or more general bounds.
The repeated three-prime question can therefore be written in three equivalent forms.

Problem 4.3 (the repeated three-prime target)**.** Prove that $S = \mathcal{R}_{\{2,3,5\}}$ is irrational.

Problem 4.4 (no reduced tail is an integer)**.** Prove that for every $a \geq 1$ and every integer $B \geq 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \tag{9}$$

Problem 4.5 (residue inequalities after every starting index)**.** Prove (8).

The equivalence follows from (5), denominator clearing and Theorem 4.2; none of the three assertions is proved for S. The companion's discussion of the repeated sum records finite calculations and the hypotheses missing from proposed finite-difference and function-theoretic arguments.

5 Two-prime transcendence

For two primes, the distinct sum is affine in one Hecke–Mahler value; the repeated sum is quadratic in it. Fan gave the factorisation of the repeated sum and its transcendence consequence in his post of 26 June 2026 [7].

Theorem 5.1 (both two-prime sums)**.** *Let $p < q$ be distinct primes. Put $\theta = \log p / \log q$ and $A = \sum_{n\geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Let $\mathcal{R}_{p,q}$ sum the reciprocal running LCM at every positive $\{p,q\}$-smooth integer, and let $\mathcal{D}_{p,q}$ count each distinct running LCM once. Then*

$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \qquad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \tag{10}$$

Both numbers are transcendental.

The Lean proof assumes the transcendence theorem of Bugeaud and Laurent; the two identities in (10) are proved in Lean without it.

Proof. Write $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the positive powers of p contribute $A = \sum_{n\geq 0} x^n y^{m_n}$. There is one q-power strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, so the positive powers of q contribute

$$B = \sum_{n\geq 0} \delta_n x^n y^{m_n+1}, \qquad \mathcal{D}_{p,q} = A + B.$$

Absolute convergence allows an index shift. Multiplying $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n}(y-1)$ by x^{n+1} and summing gives $(A-1) - xA = x(y-1)B/y$. Therefore

$$B = \frac{p - (p-1)A}{q-1}.$$

For the repeated sum, the running LCM at $p^i q^j$ is $p^{i+\lfloor j/\theta \rfloor} q^{j+m_i}$. Its reciprocal is the product of $x^i y^{m_i}$, which depends only on i, and $y^j x^{\lfloor j/\theta \rfloor}$, which depends only on j. Absolute convergence therefore gives

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, $n = \lfloor j/\theta \rfloor$ locates q^j between p^n and p^{n+1}; hence $m_n = j - 1$ and $\delta_n = 1$. This matches the positive-j terms with those of B, while $j = 0$ contributes 1. Substitution proves (10).

Factorisation alone would not prove transcendence of the product. The two identities instead express both sums as nonconstant polynomials over $\mathbb{Q}$ in the same A. It remains to prove A transcendental. For this we use the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$. Geometric summation gives

$$A = \frac{1}{1 - x} - \frac{1 - y}{y} F_\theta(x, y). \tag{11}$$

Bugeaud and Laurent's theorem [5, Theorem 1.1] applies with intercept $\rho = 0$, $\beta = x$ and $\alpha = y$: the slope θ is irrational and lies in $(0, 1)$, x and y are nonzero algebraic numbers, $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$. Thus $F_\theta(x, y)$ is transcendental; this case $\rho = 0$ is due to Loxton and van der Poorten [6, Theorem 8, p. 40]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. □

The companion's two-prime calculation records an ordinary extension to the monoid generated by coprime integers $1 < p < q$, such as $4, 9$. That monoid need not contain all integers supported on the prime divisors of pq; multiplicatively dependent generators such as $4, 8$ require a different calculation.

For larger prime sets, the sums over pure powers $E_p = \sum_{\alpha \geq 0} \mathrm{L}(p^\alpha)^{-1}$ satisfy $\mathcal{D}_P = \sum_{p \in P} E_p - (|P| - 1)$. The companion's prime-power subseries argument treats their irrationality as a remark with an ordinary proof awaiting formalisation. The source record contains no separate priority search for this claim. Irrationality of the individual subseries would not imply irrationality of their sum. The repeated series also contains mixed products of primes, whose multiplicities change the tail estimates.

A Identities for the running LCM and multiplicities

Let p, q, r be pairwise distinct primes. Write $\mathrm{H} = \mathrm{H}_{\{p,q,r\}}$ and $\mathrm{K}(i, j, k) = \mathrm{H}(p^i q^j r^k)^{-1}$ throughout this appendix and the next. The following identities underlie the grouping of the repeated series in Section 3.

Proposition A.1 (the running least common multiple)**.** *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then the running least common multiple is given by $\mathrm{L}(x) = \mathrm{H}(x)$.*

Proof. We have $u \mid \mathrm{H}(x)$ for every supported $u \leq x$. For the reverse divisibility, take the maximal powers of p, q, r below x: they belong to the prefix and are pairwise coprime, so their product divides its LCM. □

For each prime, $x/p < p^{\lfloor \log_p x \rfloor} \le x$. Multiplying gives

$$x^3/(pqr) < \mathrm{H}(x) \le x^3. \tag{12}$$

Proposition A.2 (cells and jumps)**.** *The running LCM is constant when the three integer logarithms are constant. A jump in exactly one logarithm multiplies it by the corresponding prime. The first n positive powers of each prime, together with 1, form $3n+1$ distinct points.*

Proof. The running LCM is a product of the three maximal pure powers, which proves the constancy and jump assertions. Unique factorisation separates the $3n$ positive pure powers from one another and from 1. □

Proposition A.3 (grouping terms with the same running LCM)**.** *For a finite exponent box $\mathcal{B}$, set $F(H) = \{(i,j,k) \in \mathcal{B} : \mathrm{H}(p^i q^j r^k) = H\}$. Then*

$$\sum_{(i,j,k)\in\mathcal{B}} \mathrm{K}(i,j,k) = \sum_{H} \frac{\#F(H)}{H}. \tag{13}$$

Proof. Partition the finite exponent box by the value of the running LCM. The terms in the part $F(H)$ all equal $1/H$. □

B The three-prime kernel

The two-prime proof separates the summand by exponents. For three primes, the carry in adding two fractional parts prevents any finite separation of i from (j,k). Diagonal rescaling leaves just two entry values, 1 and $1/r$; choosing their threshold positions gives nonzero minors of every order. This concerns exact kernel identities, not identities after summation.

Example B.1. For $(p,q,r) = (2,3,5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

A sum of d separated products has rank at most d in each layer. Thus the two-by-two example excludes only rank one; excluding every finite d requires nonzero minors of arbitrary order. Their indices may depend on the order, but not on k.

Theorem B.2 (no finite separation of the kernel)**.** *Let p,q,r be primes with $p \ne q$, $p \ne r$ and $q \ne r$. For every $n \ge 0$ there are injective maps $I, J : \{0,\dots,n-1\} \to \mathbb{N}$ such that, for every $k \ge 0$,*

$$\det(\mathrm{K}(I(a), J(b), k))_{0\le a,b<n} \ne 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j,k)$, $0 \le \ell < d$, satisfying

$$\mathrm{K}(i,j,k) = \sum_{\ell<d} f_\ell(i) G_\ell(j,k) \qquad \text{for all } i,j,k.$$

Proof. Fix k. The powers of p and q separate into row and column factors, as before. Only the power of r retains an interaction between i and j. Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$, where braces denote fractional parts. Then

$$\lfloor i \log_r p + j \log_r q \rfloor = \lfloor i \log_r p \rfloor + \lfloor j \log_r q \rfloor + \mathbf{1}_{\{x_i + y_j \ge 1\}}.$$

The last term is the carry when the two fractional parts are added. Putting all the other prime powers into row and column factors gives

$$\mathrm{K}(i,j,k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \qquad C_{ij} = c^{\mathbf{1}_{\{x_i+y_j\geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q (p^i r^k) \rfloor} r^{k+\lfloor \log_r p^i \rfloor},$$
$$V_j(k) = q^j p^{\lfloor \log_p (q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

The positive factors preserve nonvanishing of minors and contain all dependence on k. In column j, the entry changes from 1 to c at the threshold $1 - y_j$. Both $\log_r p$ and $\log_r q$ are irrational, so their fractional-part orbits are dense separately; no joint orbit is needed. For $n \geq 1$, choose rows with $0 < x_{I(0)} < \cdots < x_{I(n-1)} < 1$. Then choose each column threshold between consecutive row phases. Writing $s_b = 1 - y_{J(b)}$, take

$$s_0 \in (0, x_{I(0)}), \qquad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals for the s_b are disjoint, so the column indices are distinct. The strict inequalities also avoid the threshold itself, and give $x_{I(a)} + y_{J(b)} \geq 1$ exactly when $b \leq a$. Thus multiplying row a by $U_{I(a)}(k)$ and column b by $V_{J(b)}(k)$ reduces the selected kernel minor to

$$T_n(c) = \begin{pmatrix} c & 1 & \cdots & 1 \\ c & c & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \cdots & c \end{pmatrix}, \qquad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtract each preceding row from the next, working upwards from the last row. Each row difference has just one nonzero entry, $c - 1$, in its own column. The resulting triangular matrix has diagonal entries $c, c-1, \ldots, c-1$. The original determinant is therefore

$$c(c-1)^{n-1} \prod_{a<n} U_{I(a)}(k)^{-1} \prod_{b<n} V_{J(b)}(k)^{-1} \neq 0.$$

The indices were chosen from C, independently of k. For $n = 0$ the empty determinant equals 1.

For the final assertion, suppose a separation with d summands existed and fix any k. On the rows $I(0), \ldots, I(d)$ and columns $J(0), \ldots, J(d)$ its matrix would be the product of the $(d+1) \times d$ matrix $(f_\ell(I(a)))_{a,\ell}$ and the $d \times (d+1)$ matrix $(G_\ell(J(b), k))_{\ell,b}$. Its rank would be at most d, so its determinant would vanish, contrary to the minor just constructed. □

No continuity or boundedness is assumed for the separated factors.

Corollary B.3 (the same minors modulo integers coprime to 30)**.** *For $(p,q,r) = (2,3,5)$ and every $n \geq 1$, there are injective maps $I, J : \{0, \ldots, n-1\} \to \mathbb{N}$, chosen independently of B and k, such that for every $B \geq 2$ coprime to 30 and every $k \geq 0$, the selected $n \times n$ kernel matrix has unit determinant over $\mathbb{Z}/B\mathbb{Z}$, with each reciprocal prime power interpreted by its modular inverse.*

Proof. Choose the maps from Theorem B.2. Every row and column factor is a unit modulo B. The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. □

The modulus may be composite, for example 49 or 77. Coprimality with 30 makes both the reciprocal entries and the determinant units: the latter introduces only a power of 4 in its numerator.

Leading minors. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$\mathrm{K}(3, j, 0) = \frac{1}{120}\,\mathrm{K}(0, j, 0) \qquad (0 \le j < 4).$$

Evaluate the running LCM at 3^j and $8 \cdot 3^j$ to obtain these equalities. At $j = 4$, however, $\mathrm{K}(3, 4, 0) - \mathrm{K}(0, 4, 0)/120 = -1/19440000$. The selection in the theorem avoids this problem by choosing column thresholds at distinct cuts among the row phases. An arbitrary sample may miss gaps; its rank counts distinct cuts, with one exception.

Proposition B.4 (rank of a matrix of threshold columns)**.** *Let $m \ge 1$ and let c lie in a field with $c \ne 0, 1$. For $0 \le k \le m$ write v_k for the length-m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0,m\} \subseteq E}$.*

The exception is the pair of constant columns, $v_0 = c v_m$. All other distinct cuts contribute independently.

Proof. List $E = \{k_1 < \cdots < k_t\}$. The $t - 1$ differences $v_{k_{j+1}} - v_{k_j} = (1 - c)\mathbf{1}_{\{k_j, \ldots, k_{j+1} - 1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t. If $k_1 = 0$ and $k_t = m$, then $v_0 = c\,\mathbf{1}$ lies in the span of the differences (their sum is $(1 - c)\mathbf{1}$), so the rank is $t - 1$. □

Order the sampled row phases and take $c = 1/r$ over $\mathbb{Q}$. The formula then gives the rank of every nonempty rectangular sample, independently of k: the layer changes only the nonzero diagonal factors. The companion's kernel analysis gives the exact rational comparisons.

Approximation. The diagonal rescaling preserves rank but removes the kernel's decay. Retaining rows $i < N$ in the original kernel gives N separated terms; $\mathrm{H}(x) > x^3/(pqr)$ makes the error tend to zero uniformly and in $\ell^1(\mathbb{N}^3)$. The rescaled matrix instead has uniform distance $(1 - 1/r)/2$ from finite-rank matrices. The companion's approximation theorem proves both statements. Neither implies irrationality of the scalar sum.

C Verification, reproducibility and sources

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise

correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The five-map proof is in the distinct-value development and its block-series criterion. Its statement has no recorded Comparator comparison. The evidence record fixes the source versions; the named input for two-prime transcendence is stated beside that theorem.

The programs for sums over distinct values calculate the interval endpoints and finite-state examples using exact arithmetic; the proof in Section 2 derives its constants directly.

The companion compares one-jump and dyadic bounds. Its finite-state extension uses earlier blocks to sharpen the intervals. Saved exact computations satisfy the criterion for every three-prime set and 292 of the 330 four-prime sets of primes at most 31. The criterion and computations remain remarks awaiting formalisation. For $\{2,3,5,7\}$, two admissible automaton words induce the same affine map, defeating first-block recovery at every refinement. Equal maps alone do not give equal arithmetic tails.

The companion's source inventory covers the ordinary general and subseries arguments, the 26 September 2026 AI check and its corrections. No independent human mathematical review has been reported.

Further comparisons

The companion compares our bounded-base series with the criteria of Erdős–Straus [9], Hančl–Tijdeman [10] and Diananda–Oppenheim [8]; the bounded-ratio argument with Erdős–Taylor [11, Theorem 1, p. 600] and Fan [12, Lemma 3.1, p. 7]; and integer recurrences with Hančl–Tijdeman's polynomial Cantor-series results [13, Theorems 2.2, 3.1 and 4.2]. Koutsoukou-Argyraki and Li's Isabelle development [19] formalises classical criteria, not this repeated series.

The transcendence comparisons concern finite differences [14], echoing [15], fixed-base digit complexity [20] and multivariate Mahler theory [21]. The companion exhibits nonzero third-difference boundary terms and distinguishes variable-base digits from fixed-base expansions: quadratic growth supplies neither the missing cancellation nor recurrence hypotheses. Fixed-prime semigroups [16, 17] and other Ahmes-series problems [18] supply context, not estimates used in our counting proof.

Statements and declarations

Authorship and availability. AI agents carried out most of the research and drafting. The companion contains the longer proofs and exact computations, together with the hypotheses and limitations of the unsuccessful approaches.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator pending	Lemma 3.1	Lean · Comparator
Proposition 3.2	Lean · Comparator	Theorem 3.3	Lean · Comparator
Corollary 3.4	Lean · Comparator	Lemma 4.1	Lean · Comparator
Theorem 4.2	Lean · Comparator	Theorem 5.1	Lean[†]
Proposition A.1	Lean · Comparator	Proposition A.2	Lean · Comparator
Proposition A.3	Lean · Comparator	Theorem B.2	Lean · Comparator
Corollary B.3	Lean · Comparator	Proposition B.4	Lean · Comparator